

National Institute of Technology, Rourkela
End Semester Examination, Autumn 2021
Compiler Design (CS3007)

5th semester B.Tech in Computer Science & Engineering

Duration: 2 Hours
Number of pages: 2

Full Marks: 50

- Attempt all questions.
- All parts of a question should be answered together.
- Consider ρ as last four digits of your roll number, for example if your roll number is 117CS0123 then $\rho = 0123$.

- Q 1. (a) Construct a syntax directed translation (SDT) scheme to convert a decimal number to hexadecimal. [5]
(b) Give a trace of converting ρ in decimal to hexadecimal using the SDT scheme. [5]

Q 2. Consider the following programs:

<pre>(i) int * A(); void main() { int * p; p=A(); } int * A() { int a=5; return &a; }</pre>	<pre>(ii) void main() { float t=1.25f; printf((t==1.25)?"Yes":"No"); } (iii) void main() { int a=5, b=0, c; c=a/b; }</pre>
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- (a) Find the syntax, semantic and runtime errors. [4]
(b) Explain their outcome with causes. [3]
(c) How to prevent them? [3]

Q 3. Rewrite the following code fragments using appropriate optimization techniques. [5×2]

<pre>(a) a=5; x=2*a; y=x+6; t=x*y;</pre>	<pre>(b) int f(int x,int y){ int z=x*y; return x+y; return x-y; }</pre>	<pre>(c) a=b+c; b=a-d; c=b+c; d=a-b;</pre>
<pre>(d) if(a<b) z=x*2; else y=10; g=x*2;</pre>	<pre>(e) for(k=0;k<100;k=k+1) for(j=0;j<100;j=j+1) for(i=0;i<5000;i=i+1) x[i][j]=2*x[i][j];</pre>	

Provide name of the applied optimization technique. Explain the improvement achieved by the optimized code.

- Q 4. (a) Specify the storage of variables and their contents during execution of following program. [5]
- ```
1.#include<stdlib.h>
2.int a=5;
3.int main()
4. {
5. float f=0.2f;
6. static int j=2;
7. char *s;
8. s=(char*)malloc(sizeof(char)*5);
9. return 0;
10. }
```

- (b) Provide activation records for the trace of following program.

```

1. #include <stdio.h>
2. int fact(int);
3. int main()
4. {
5. int r;
6. r=fact(5);
7. printf("%d",r);
8. return 0;
9. }
10. int fact(int n)
11. {
12. int p;
13. if(n<0)
14. {
15. printf("Error!");
16. return -1;
17. }
18. if((n==0)|| (n==1))
19. return 1;
20. p=n*fact(n-1);
21. return p;
22. }

```

The format of activation record for procedure call is as follows

|                      |
|----------------------|
| Actual parameters    |
| Return values        |
| Control link         |
| Saved machine status |
| Local data           |

- Q 5. The description of source language operators and target machine are given as follows

Source language description

| <i>op_list</i> | 0 | 1 | 2 | 3 | 4 | Precedence<br>Highest | Operator<br>* / %<br>+ - | Associativity<br>left to right<br>left to right |
|----------------|---|---|---|---|---|-----------------------|--------------------------|-------------------------------------------------|
|                | + | - | * | / | % |                       |                          |                                                 |

Target machine description

Register bank,  $R = \{r_0, r_1, r_2, r_3\}$

Memory locations,  $M = \{m_0, m_1, \dots\}$

Instruction Set

MOV  $X, Y; X \leftarrow Y, ((X \in R) \wedge (Y \in R \cup M)) \vee ((X \in R \cup M) \wedge (Y \in R))$

ADD  $X, Y; X \leftarrow X + Y, (X \in R) \wedge (Y \in R \cup M)$

SUB  $X, Y; X \leftarrow X - Y, (X \in R) \wedge (Y \in R \cup M)$

MUL  $X, Y; X \leftarrow X * Y, (X \in R) \wedge (Y \in R \cup M)$

DIV  $X, Y; X \leftarrow X / Y, (X \in R) \wedge (Y \in R \cup M)$

MOD  $X, Y; X \leftarrow X \% Y, (X \in R) \wedge (Y \in R \cup M)$

- (a) Write the three address code for the expression  $A \text{ op}_1 B \text{ op}_2 C \text{ op}_3 D \text{ op}_4 E$

[5]

- identifiers  $A, B, C, D$  and  $E$  are integer type variables stored in memory
- operator  $\text{op}_i = \text{op\_list}[\rho\%(i+1)], 1 \leq i \leq 4$

- (b) Generate the target code for the expression using labeled tree method.

[5]

\*\*\*\*\*All the best\*\*\*\*\*